Production Overview

Total Unit Produced

3,298,062

Average Daily OEE

61.21%

Unplanned Downtime Hours

171.31

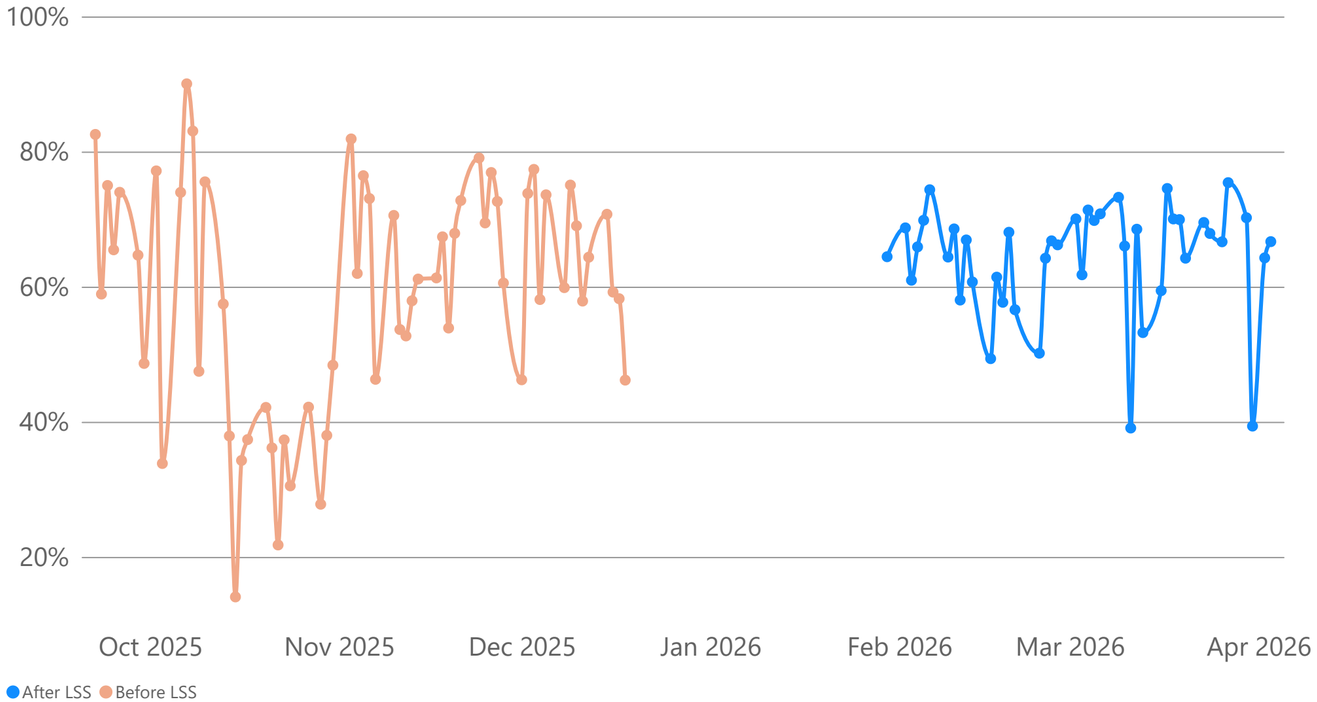
Average Daily Quality

97.82%

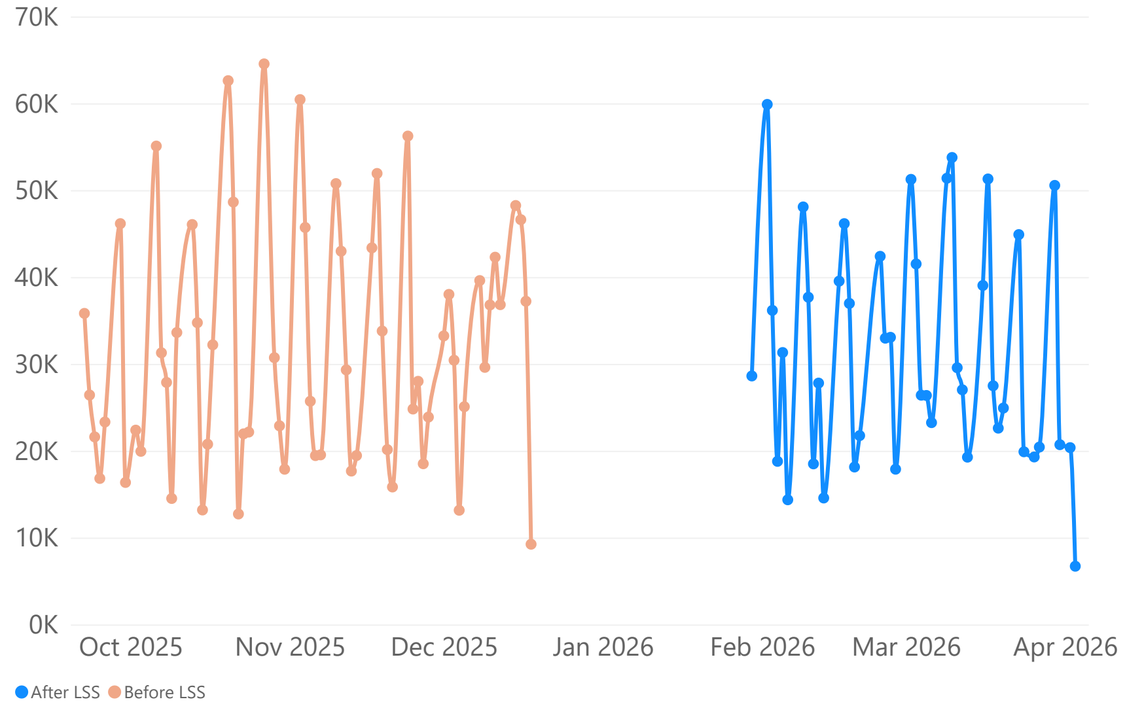
Period Slicer

- Select all
- After LSS
- Before LSS

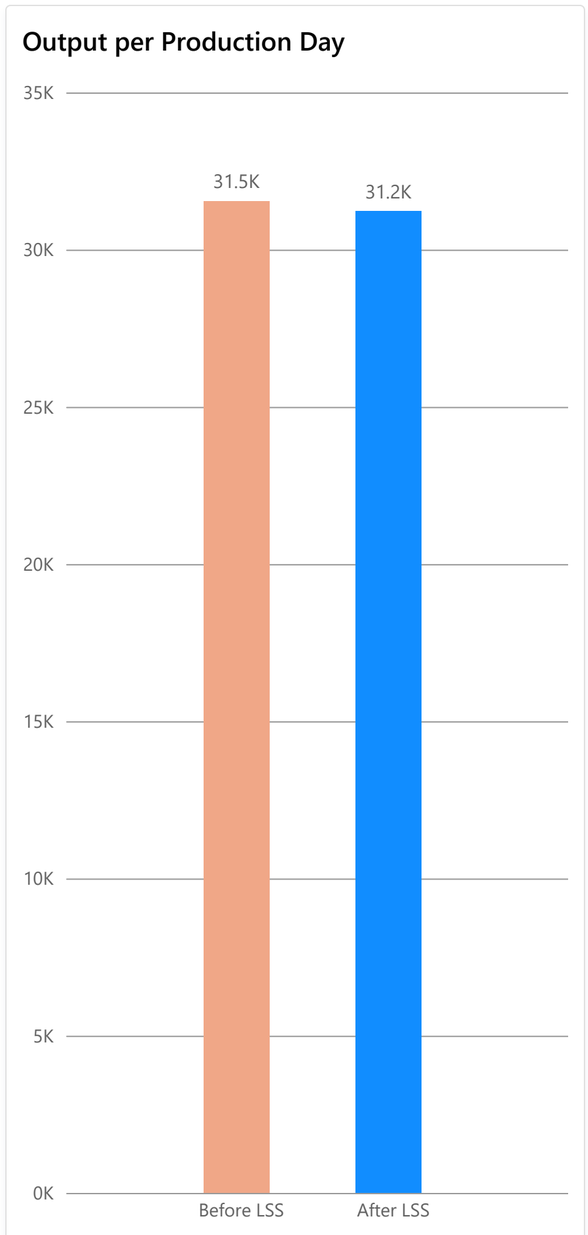
Daily OEE Trend

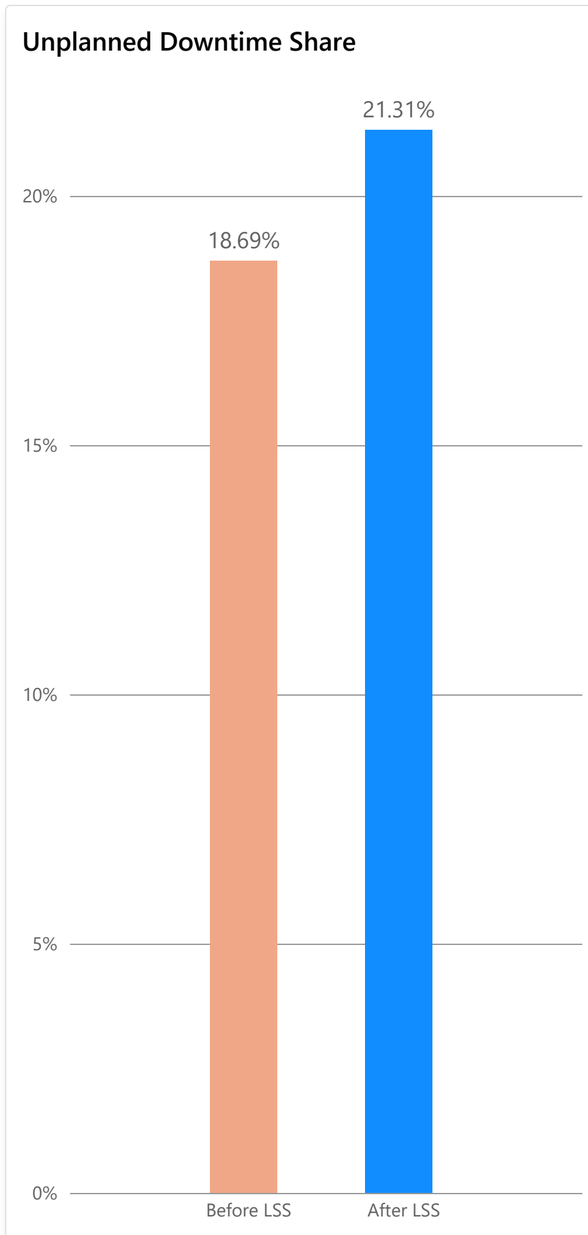


Daily Output



Before And After LSS





Quality Rate

Before LSS	98.50%
After LSS	96.82%

Availability Rate

Before LSS	59.17%
After LSS	69.66%

Performance Rate

Before LSS	93.20%
After LSS	93.74%

Time-Weighted Reported OEE

Before LSS	54.36%
After LSS	63.99%

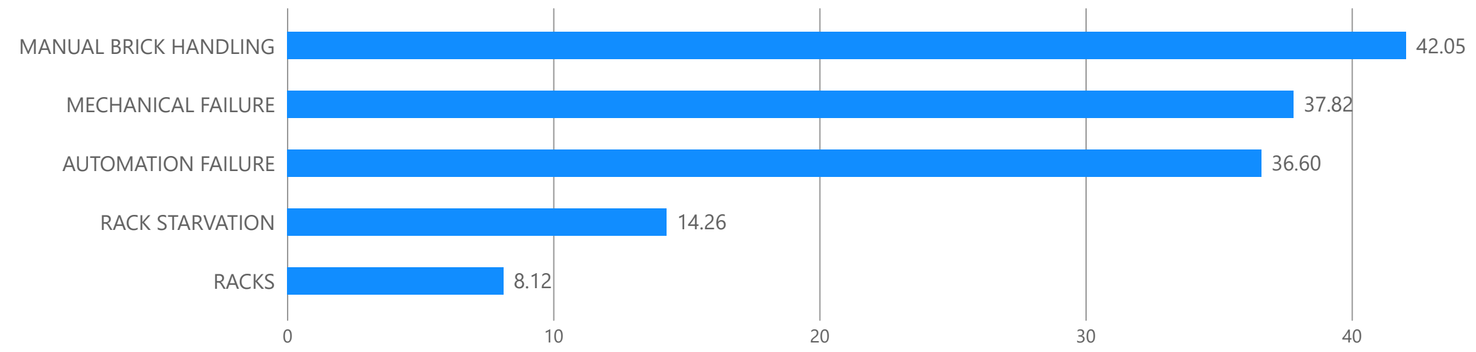
Before LSS: 62 observed production days; After LSS 43.

Downtime Analysis

Period

- Select all
- After LSS
- Before LSS

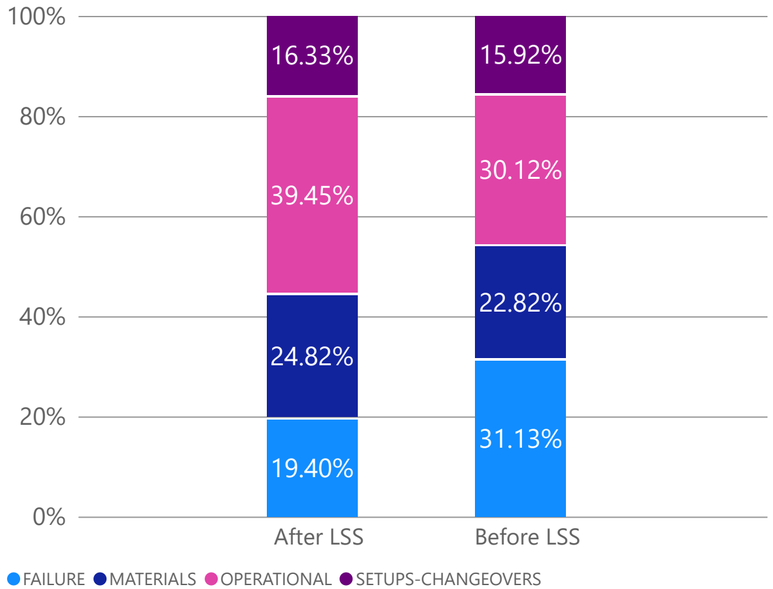
Recorded Stop Hours by Stop



Recorded Stop Hours

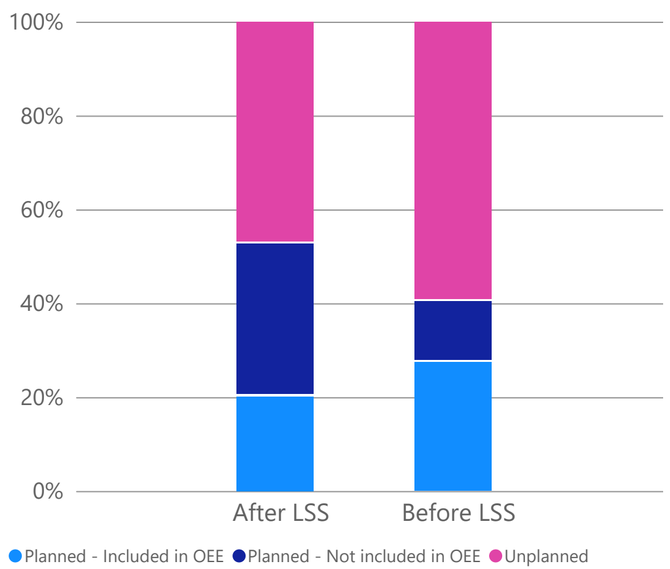
Before LSS	173.13
After LSS	145.17

Stop Group Share by Period and Stop group



Stop Type Time Share

by Period and Stop type



Average Stop Time Duration (min)

Before ...	16.36
After LSS	16.13

OEE summaries: 62 dates before LSS, 43 after.
Stop-event logs: 62 dates before, 45 after.

Sources And Methodology

Source

Independent portfolio project by Caio Bergamo using the OEE and downtime datasets published by Panos Ntoas on Zenodo. The source describes a continuous brick manufacturing line. License: Creative Commons Attribution 4.0 International (CC BY 4.0), as stated in the dataset README. zenodo.org

Dataset: https://doi.org/10.5281/zenodo.17855209

Metric Definitions

Time-weighted reported OEE: supplied daily OEE weighted by planned production seconds.

- **Availability:** full-speed plus slow-production seconds ÷ planned production seconds.
- **Performance:** total output ÷ ideal production quantity.
- **Quality:** good units ÷ total output.
- **Output per production day:** total output ÷ distinct observed production dates.
- **Unplanned downtime share:** unplanned stop seconds ÷ planned production seconds.
- **Stop group/type time shares:** recorded category duration ÷ total recorded stop duration within each period.
- **Hours:** OEE time fields ÷ 3,600; recorded stop minutes ÷ 60.

Scope & Preparation

Before Lean Six Sigma (LSS): September 22–December 18, 2025; 62 production-date records and 635 stop events.
After LSS: January 30–April 3, 2026; 43 production-date records and 540 stop events.

Prepared in Power Query: removed empty rows and columns, checked data types, standardized column names, added period labels, and appended matching datasets. Greek operator comments were excluded from analysis. Original source files were preserved.

Validation & Limitations

Retained values and displayed summary metrics were checked against the original CSVs. Dates were unique in both OEE files. Calculated event durations were positive; recorded durations differed by up to one minute. Original durations and unspecified locations were retained.

Eleven baseline reported OEE values differ from Availability x Performance x Quality and approximately match Availability x Performance. Original OEE was retained. Separately, 2 baseline and 9 after-LSS Performance values exceed 100%. Ideal production quantity is interpreted as units, despite the source README's seconds label.

Unplanned hours differ between OEE summaries and event logs: 103.00 vs 103.09 before; 68.31 vs 68.68 after. OEE tables cover 62/43 dates; stop logs cover 62/45. Periods have unequal lengths and do not establish that LSS caused the changes. Stop-time shares describe composition.